

# FPROS6-A790-OSB

Serie FPROS • R513A • senza serbatoio / without tank



FPROS6-A790-OSB

RESA FRIGORIFERA • COOLING CAPACITY

**517** kW

EER • 12/7 °C • AIR 35 °C

**3,00**

RESA NOMINALE • 15 °C • AIR 25 °C

**789** kW

PORTATA ACQUA • WATER FLOW

**88,9** m<sup>3</sup>/h

|                            |                 |                                        |                       |
|----------------------------|-----------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 7.015 × 2.197 × 2.442 |
| Circuiti / Circuits        | 1               | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 12              | Taglia / Size                          | 790                   |
| Free cooling (ΔT 8 K)      | 500 kW *        | Campo aria esterna / Ambient air range | -5 / +47 °C           |
| Refrigerante / Refrigerant | R513A • GWP 631 |                                        |                       |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 578   | 548   | 517   | 486   | 454   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 789   | 749   | 708   | 668   | 626   |

Refrigeratori con free cooling integrato e compressori a vite, per grandi potenze che lavorano tutto l'anno.

Chillers with integrated free cooling and screw compressors, for large capacities running all year round.

# Dati tecnici

## Technical data

FPROS6 - A790 - OSB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 788,78 kW                |
| Resa frigorifera / Cooling capacity | 678.351 kcal/h           |
| Potenza assorbita / Power input     | 147,09 kW                |
| EER / EER                           | 5,36                     |
| Portata acqua / Water flow          | 135,67 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                         |
|-------------------------------------|-------------------------|
| Resa frigorifera / Cooling capacity | 516,80 kW               |
| Potenza assorbita / Power input     | 172,29 kW               |
| EER / EER                           | 3,00                    |
| ESEER / ESEER                       | 3,96                    |
| Portata acqua / Water flow          | 88,89 m <sup>3</sup> /h |

### Componenti / Components

|                                               |                      |
|-----------------------------------------------|----------------------|
| Compressori / Compressors                     | 2 × Vite             |
| Circuiti frigoriferi / Refrigerant circuits   | 1                    |
| Ventilatori / Fans                            | 12                   |
| Free cooling (ΔT 8 K) / Free cooling (8 K ΔT) | 500 kW *             |
| Ventilatori free cooler / Free cooler fans    | 12                   |
| Condensatore / Condenser                      | a microcanali        |
| Evaporatore / Evaporator                      | a fascio tubiero     |
| Refrigerante / Refrigerant                    | R513A · GWP 631 · A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 7.015 × 2.197 × 2.442 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +47 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

\* Potenza di free cooling indicativa, da confermare in fase di offerta. / Indicative free cooling capacity, to be confirmed at quotation stage.

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 · 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 506,5 kW     |
| MEG 30 % | × 0,950 | 491,0 kW     |
| MEG 40 % | × 0,920 | 475,5 kW     |
| MPG 20 % | × 0,975 | 503,9 kW     |
| MPG 30 % | × 0,960 | 496,1 kW     |
| MPG 40 % | × 0,950 | 491,0 kW     |

# Prestazioni Performance

FPROS6-A790-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 499 | 475 | 451 | 427 | 401 | 376 | 350 |
| 2             | 520 | 496 | 471 | 446 | 419 | 393 | 366 |
| 3             | 542 | 517 | 491 | 464 | 437 | 410 | 382 |
| 4             | 565 | 539 | 512 | 485 | 457 | 428 | 400 |
| 5             | 588 | 561 | 533 | 505 | 476 | 447 | 417 |
| 6             | 612 | 584 | 556 | 526 | 496 | 466 | 435 |
| 7             | 637 | 608 | 578 | 548 | 517 | 486 | 454 |
| 8             | 663 | 633 | 602 | 571 | 539 | 506 | 473 |
| 9             | 689 | 658 | 626 | 594 | 560 | 527 | 493 |
| 10            | 717 | 685 | 652 | 618 | 584 | 549 | 514 |
| 11            | 744 | 711 | 677 | 642 | 607 | 571 | 534 |
| 12            | 774 | 740 | 704 | 668 | 631 | 594 | 557 |
| 13            | 803 | 768 | 731 | 694 | 656 | 618 | 579 |
| 14            | 834 | 798 | 760 | 722 | 682 | 643 | 602 |
| 15            | 865 | 828 | 789 | 749 | 708 | 668 | 626 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 118 | 129 | 141 | 154 | 169 | 186 | 204 |
| 2             | 119 | 129 | 142 | 155 | 170 | 186 | 204 |
| 3             | 119 | 130 | 142 | 155 | 170 | 187 | 205 |
| 4             | 119 | 130 | 142 | 156 | 171 | 187 | 206 |
| 5             | 120 | 131 | 143 | 156 | 171 | 188 | 206 |
| 6             | 120 | 131 | 143 | 157 | 172 | 188 | 207 |
| 7             | 121 | 132 | 144 | 157 | 172 | 189 | 207 |
| 8             | 121 | 132 | 144 | 158 | 173 | 190 | 208 |
| 9             | 121 | 132 | 144 | 158 | 173 | 190 | 209 |
| 10            | 122 | 133 | 145 | 158 | 174 | 190 | 209 |
| 11            | 122 | 133 | 145 | 159 | 174 | 191 | 210 |
| 12            | 123 | 134 | 146 | 159 | 174 | 191 | 210 |
| 13            | 123 | 134 | 146 | 160 | 175 | 192 | 211 |
| 14            | 124 | 135 | 147 | 160 | 175 | 192 | 211 |
| 15            | 124 | 135 | 147 | 160 | 176 | 193 | 211 |

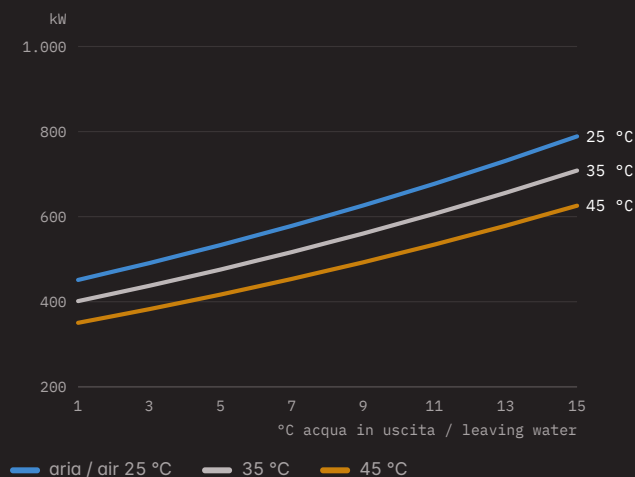
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   |
|---------------|------|------|------|------|------|------|------|
| 1             | 4,22 | 3,68 | 3,20 | 2,76 | 2,37 | 2,03 | 1,72 |
| 2             | 4,39 | 3,83 | 3,33 | 2,88 | 2,47 | 2,11 | 1,79 |
| 3             | 4,56 | 3,98 | 3,46 | 2,99 | 2,57 | 2,19 | 1,87 |
| 4             | 4,73 | 4,14 | 3,60 | 3,11 | 2,67 | 2,29 | 1,94 |
| 5             | 4,91 | 4,29 | 3,73 | 3,23 | 2,78 | 2,38 | 2,02 |
| 6             | 5,10 | 4,46 | 3,88 | 3,36 | 2,89 | 2,47 | 2,10 |
| 7             | 5,28 | 4,62 | 4,03 | 3,49 | 3,00 | 2,57 | 2,19 |
| 8             | 5,48 | 4,80 | 4,18 | 3,62 | 3,12 | 2,67 | 2,27 |
| 9             | 5,67 | 4,97 | 4,34 | 3,76 | 3,24 | 2,77 | 2,36 |
| 10            | 5,88 | 5,16 | 4,50 | 3,90 | 3,36 | 2,88 | 2,46 |
| 11            | 6,08 | 5,34 | 4,66 | 4,04 | 3,49 | 2,99 | 2,55 |
| 12            | 6,30 | 5,53 | 4,83 | 4,20 | 3,62 | 3,11 | 2,65 |
| 13            | 6,51 | 5,73 | 5,00 | 4,35 | 3,75 | 3,22 | 2,75 |
| 14            | 6,74 | 5,93 | 5,18 | 4,51 | 3,89 | 3,34 | 2,85 |
| 15            | 6,96 | 6,13 | 5,36 | 4,67 | 4,03 | 3,46 | 2,96 |

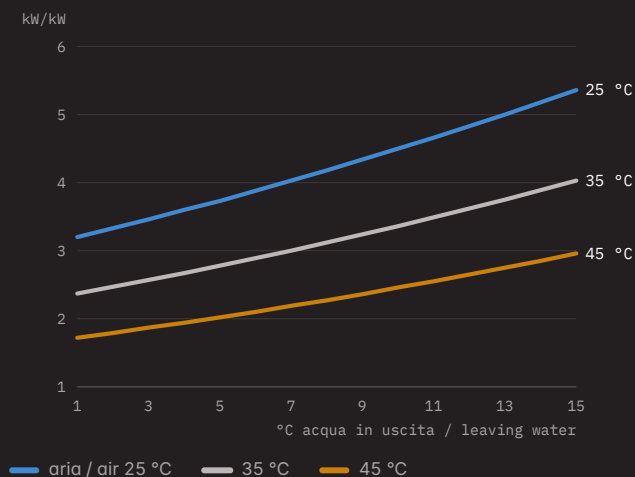
# Curve e dimensioni

## Curves and dimensions

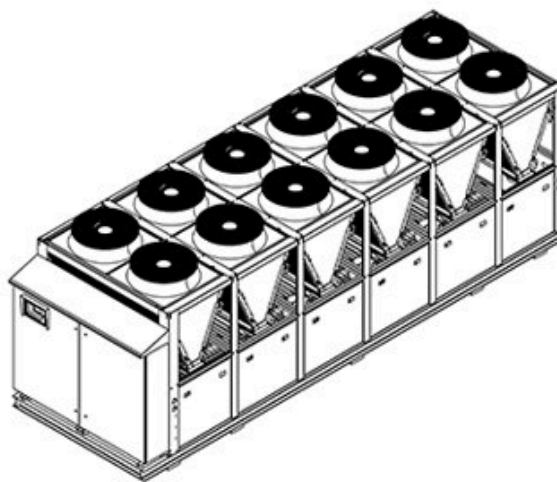
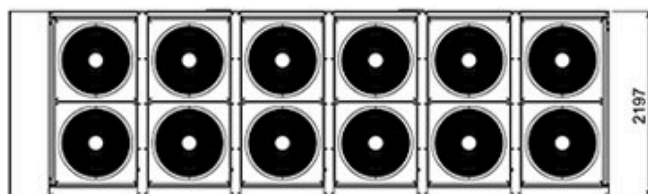
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 7.015 × 2.197 × 2.442 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

FPROS6-A790-OSB · 65 optional disponibili per questa configurazione · 65 options available for this configuration

## Pompe / Pumps

Pompa 92 m³/h – prevalenza 25 m

Pump 92 m³/h – head 25 m

Pompa 92 m³/h – prevalenza 40 m

Pump 92 m³/h – head 40 m

Pompa doppia montata 92 m³/h – prevalenza 25 m

Twin pump, factory-fitted, 92 m³/h – head 25 m

Pompa doppia montata 92 m³/h – prevalenza 40 m

Twin pump, factory-fitted, 92 m³/h – head 40 m

## Serbatoi e vasi di espansione / Tanks and expansion vessels

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 2000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata

External water tank 2000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata

External water tank 3000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata

External water tank 5000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 1500 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 2000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata

External water tank 1500 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata

External water tank 2000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata

External water tank 3000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata

External water tank 5000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 2000 l, pressurised, galvanised steel with factory-fitted twin pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 3000 l, pressurised, galvanised steel with factory-fitted twin pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted twin pump

Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 1500 l, pressurised, galvanised steel with factory-fitted single pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump

Vaso di espansione 100 l

Expansion vessel 100 l

Vaso di espansione 200 l

Expansion vessel 200 l

Vaso di espansione 300 l

Expansion vessel 300 l

Vaso di espansione 500 l

Expansion vessel 500 l

Vaso di espansione 750 l

Expansion vessel 750 l

## Valvole e filtri / Valves and filters

Elettrovalvola 2"

Solenoid valve 2"

Elettrovalvola 3"

Solenoid valve 3"

Elettrovalvola 4"

Solenoid valve 4"

Elettrovalvola 5"

Solenoid valve 5"

Elettrovalvola 6"

Solenoid valve 6"

Filtro a dischi in acciaio 2"

Steel disc filter 2"

# Optional — 2/2

## Options

FPROS6 - A790 - OSB · 65 optional disponibili per questa configurazione · 65 options available for this configuration

### Valvole e filtri / Valves and filters

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y - strainer 2"

Filtro meccanico a Y 3"

Y - strainer 3"

Filtro meccanico a Y 4"

Y - strainer 4"

Filtro meccanico a Y 5"

Y - strainer 5"

Filtro meccanico a Y 6"

Y - strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

### Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza

evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
  - Free cooling: potenza riferita a una differenza di 8 K tra acqua e aria esterna; valore indicativo, da confermare in fase di offerta.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.
  - Free cooling: capacity referred to an 8 K difference between water and outdoor air; indicative value, to be confirmed at quotation stage.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

Procryon — Smart Cooling Engineering S.r.l.  
Via Ugo Vaglia 10 — Gavardo (BS) — Italia  
[www.procryon.com](http://www.procryon.com)



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Continuity by design